DTC P1336:00 [SKYACTIV-D 1.5]

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DTC P1336:00	CKP sensor/CMP sensor malfunction
DETECTION CONDITION	• If the phase difference between the CKP sensor and CMP sensor exceeds the specified value, the PCM determines that the timing chain is stretched and stores DTC P1336:00. **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• CKP sensor malfunction — CKP sensor connector or terminals malfunction — CKP sensor is dirty — CKP sensor pulse wheel malfunction • CMP sensor malfunction — CMP sensor connector or terminals malfunction — CMP sensor is dirty — Damage to detection area of CMP sensor • PCM connector or terminals malfunction • Improper valve timing • Timing chain is stretched.
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**INSPECT CKP SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the CKP sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 13.
		No	Go to the next step.
4	**INSPECT CKP SENSOR FOR ANY FOREIGN MATTER** • Remove the CKP sensor. (See CRANKSHAFT POSITION (CKP) SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • Inspect the CKP sensor for any foreign matter. • Is there any foreign matter in the CKP sensor?	Yes	Remove the foreign matter from the CKP sensor, then go to Step 13.
		No	Go to the next step.
5	**INSPECT CKP SENSOR PULSE WHEEL** • Visually inspect the CKP sensor pulse wheel. • Are there any damage or scratches at the CKP sensor pulse wheel?	Yes	Replace the CKP sensor pulse wheel, then go to Step 13.
		No	Go to the next step.

STEP	INSPECTION		ACTION
6	**INSPECT CKP SENSOR** • Inspect the CKP sensor. (See CRANKSHAFT POSITION (CKP) SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the CKP sensor, then go to Step 13. (See CRANKSHAFT POSITION (CKP) SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
7	**INSPECT CMP SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the CMP sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 13.
		No	Go to the next step.
8	**INSPECT CMP SENSOR FOR ANY FOREIGN MATTER** • Remove the CMP sensor. (See CAMSHAFT POSITION (CMP) SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • Inspect the CMP sensor for any foreign matter. • Is there any foreign matter in the CMP sensor?	Yes	Remove the foreign matter from the CMP sensor, then go to Step 13.
		No	Go to the next step.
9	**INSPECT CMP SENSOR PULSE WHEEL** • Visually inspect the CMP sensor pulse wheel. • Are there any damage or scratches at the CMP sensor pulse wheel?	Yes	Replace the CMP sensor pulse wheel, then go to Step 13. (See CYLINDER HEAD GASKET REPLACEMENT [SKYACTIV-D 1.5].)
		No	Go to the next step.
10	**INSPECT CMP SENSOR** • Inspect the CMP sensor. (See CAMSHAFT POSITION (CMP) SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the CMP sensor, then go to Step 13. (See CAMSHAFT POSITION (CMP) SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
11	**INSPECT PCM CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 13.
		No	Go to the next step.
12	**INSPECT VALVE TIMING** • Inspect the valve timing. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Adjust the valve timing to the correct timing, then go to the next step. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Replace the timing chain, then perform the following procedure: (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • Perform the “FUEL INJECTOR INJECTION AMOUNT CORRECTION” procedure. (See FUEL INJECTOR INJECTION AMOUNT CORRECTION [SKYACTIV-D 1.5].) • Perform the “TIMING CHAIN LEARNING” procedure. (See TIMING CHAIN LEARNING [SKYACTIV-D 1.5].) Go to the next step.
13	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Start the engine. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
14	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.